

Methodology

Every quantitative claim in this document was verified directly against source files (model.py, train_unet.py, infer.py, lines_to_masks.py, label_line.py, review_and_label.py), raw training logs (train.log through train7.log), and recorded shell history for tool invocations -- not reconstructed from memory. An earlier draft of this section contained two errors (an incorrect count of annotation rounds, and a conflated train-split/total-dataset figure); both are corrected here, with the corrections noted where relevant.

III. Methodology

A. Problem Formulation

The task addressed is the automated localization of the growth-band centerline in cross-sectional photographs of surf clam (*Spisula solidissima*) hinge sections, a measurement traditionally performed manually to support age and growth-rate estimation. Given an RGB image I of a hinge cross-section, the objective is to produce a one-dimensional curve, ordered from the hinge (umbo) to the ventral margin, that traces the visual center of the growth-band ridge structure. This is formulated as a per-pixel binary segmentation problem -- predicting a narrow band around the true centerline -- followed by a deterministic post-processing stage that collapses the predicted band into a single-pixel-wide curve.

This formulation was arrived at after two alternative formulations were implemented and found inadequate: (1) whole-shell segmentation via classical thresholding, with the centerline derived geometrically from the segmented region, and (2) whole-shell segmentation masks generated automatically via a pretrained promptable segmentation model (FastSAM) without human-verified ground truth. Both were found to fail specifically on specimens where debris or fragment material was fused to the shell in the photograph, motivating the direct-centerline-supervision approach described below.

B. Dataset and Annotation Protocol

Source imagery was drawn from the NOAA Surf Clam Survey archive, comprising RGB photographs of hinge cross-sections at native resolutions ranging from approximately 2100 to 6700 pixels in width, captured across multiple survey years (1986, 2008, 2011-2016) plus a curated 'reader comparison' subset. The full accessible pool across these sources totals 815 candidate images of this photographic type.

Ground-truth annotation was performed using two purpose-built interactive tools rather than a general-purpose annotation platform:

1. Direct annotation: for images with no existing model prediction, the annotator placed a sequence of point clicks tracing the centerline from the hinge to the ventral tail, typically 6-15 points per image.
2. Prediction-assisted review: for subsequent rounds, the current model's prediction was rendered on the image prior to annotation. The annotator either accepted the prediction unmodified or supplied a corrective sequence of clicks, which replaced the model output as ground truth. Predicted segments for which the model produced no confident output were visually distinguished from confident segments, directing annotator attention toward the regions of highest correction value.

At the time of the evaluation reported in Section H, the annotated set comprised 107 images: 37 from an initial direct-annotation phase (drawn from the curated reader-comparison subset), followed by two prediction-assisted rounds contributing 40 and 30 images respectively (77 and 107 cumulative total). The first prediction-assisted round used uniform random sampling from the unannotated pool; the second used a coverage-prioritized strategy, in which a larger candidate pool (150 images) was pre-scored by the fraction of the classically-estimated shell extent for which the current model produced confident output, with the lowest-coverage candidates selected for annotation first. Both the random and coverage-prioritized sampling modes are implemented as selectable options in the same tool; use of the coverage-prioritized mode for the second round is confirmed by the recorded command invocation.

Correction from an earlier draft: this section previously stated 'three subsequent prediction-assisted rounds.' Cross-checking training-log headers (train6.log: 77 total images; train7.log: 107 total images) against recorded tool invocations confirms exactly two such rounds (contributing 40 and 30 images), not three.

C. Ground-Truth Target Construction

Point-based annotations were converted into a rasterized supervision target via cubic B-spline interpolation (`scipy.interpolate.splprep/splev`; degree $k = \min(3, n-1)$ for n annotated points; smoothing factor $s=0$) through the annotated points, resampled to 300 curve points. A binary band mask was then rasterized by drawing this curve as a polyline of thickness $2w$, where the half-width $w = 0.02 \times \max(H, W)$ is set proportionally to the longer image dimension. Both the source image and the resulting mask were resized with aspect-ratio preservation and zero-padded to a fixed 512x512 resolution prior to network training.

D. Network Architecture

The centerline predictor is a convolutional encoder-decoder network following the U-Net design [1], comprising four downsampling stages, a bottleneck, and four upsampling stages connected via skip connections between corresponding-resolution encoder and decoder features. Each stage consists of two 3x3 convolutions (same-padding) with batch normalization and ReLU activation. Channel width follows the standard doubling convention (16, 32, 64, 128, 256 at the bottleneck), and spatial downsampling/upsampling is performed via 2x2 max-pooling and 2x2-stride transposed convolution, respectively. The network takes a 512x512x3 input and produces a 512x512x1 output map of unnormalized logits, with no output activation applied within the network itself. The complete network comprises 1,944,049 trainable parameters, verified by direct enumeration; no pretrained weights are used at any stage -- the model is trained from random initialization.

Full layer-by-layer architecture, verified tensor shapes at every stage, and a parameter-count breakdown by component are provided in the companion document, 'U-Net Architecture'.

E. Training Procedure

The network was trained to minimize a compound objective combining pixel-wise binary cross-entropy and a soft Dice coefficient loss:

$$L = L_{\text{BCE}}(\text{sigmoid}(\text{logits}), \text{target} ; \lambda) + L_{\text{Dice}}(\text{sigmoid}(\text{logits}), \text{target})$$

Because target pixels constitute a minority class (empirically 3.3-3.5% of pixels across training images, depending on the round), the positive class was upweighted in the BCE term by a factor

$$\lambda = \sqrt{(1 - p) / p}$$

where p is the measured foreground pixel fraction on the training split for the given run (yielding λ in the range 5.3-5.4 across the rounds using this weighting). This weighting was arrived at empirically: an initial experiment using the full (non-square-rooted) inverse-frequency weight ($\lambda = 27.9$) was found, by direct comparison of model predictions against held-out ground truth, to induce systematic over-prediction -- thick, spatially offset mask regions rather than a precise centerline -- consistent with the loss function's asymmetric penalty structure making false negatives on the rare class disproportionately costly relative to false positives on the majority class.

Optimization used Adam (learning rate $1e-3$) with cosine-annealed learning rate decay over 150 epochs, batch size 4. A random 15% of the annotated set was held out for validation at each training run (reshuffled, fixed random seed, when the annotated set size changed between rounds); the checkpoint corresponding to the lowest validation loss across the run was retained. Training-time data augmentation comprised random horizontal and vertical flips ($p=0.5$ each), random rotation by a multiple of 90 degrees (uniform over four orientations), and multiplicative/additive brightness jitter (gain $\sim \text{Uniform}(0.8, 1.2)$, bias $\sim \text{Uniform}(-20, 20)$, applied with $p=0.5$), applied identically to image and mask for geometric transforms and to the image only for photometric jitter.

Run	Total images (train+val)	Target	Loss weighting
train.log	38 (33+5)	whole-shell (abandoned)	unweighted
train2.log	34 (29+5)	whole-shell (abandoned)	unweighted
train3.log	37 (32+5)	centerline band	unweighted
train4.log	37 (32+5)	centerline band	$\lambda=27.9$
train5.log	37 (32+5)	centerline band	$\lambda=5.3$
train6.log	77 (66+11)	centerline band	$\lambda=5.4$

train7.log	107 (91+16)	centerline band	lambda=5.3 (current)
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F. Iterative (Human-in-the-Loop) Training Set Expansion

Because initial models trained on the 37-image direct-annotation set exhibited two identifiable failure modes -- incomplete prediction coverage over portions of the shell, and (after correcting for the former via loss reweighting) systematic centerline mislocalization on a subset of images -- model improvement was pursued through an iterative annotation-training cycle rather than through architecture modification alone.

Five training runs were conducted in total for the direct-centerline-supervision approach. The first three used the fixed 37-image annotated set and differed only in loss weighting (Section E): unweighted, then $\lambda=27.9$, then $\lambda=5.3$, motivated by the failure modes described above. The remaining two runs followed each prediction-assisted annotation round (Section B), using the resulting 77-image and 107-image sets respectively. This process is characterized as a form of human-in-the-loop active learning. Sample-efficiency gains from coverage-prioritized versus random candidate selection were not isolated or measured against each other.

Correction from an earlier draft: this section previously described 'four cycles... expanding the annotated set from 32 to 37... to 77 to 107,' which incorrectly presented 32 (a train-split count) as a sequential total-dataset size. 32 was never a total annotated-set size; it is the training portion of the 37-image set (32 train + 5 validation = 37 total).

G. Inference and Post-Processing

At inference, the trained network is evaluated on a letterboxed 512x512 representation of the input image, and the output logit map is resized (bilinear interpolation) back to native resolution after sigmoid activation, yielding a dense per-pixel confidence map $p(x,y)$ in $[0,1]$. A binary mask is obtained by thresholding at $p > 0.5$, retaining only the largest 8-connected component and filling enclosed holes. Rather than deriving the centerline as the unweighted vertical midpoint of this binary mask per image column -- a formulation found to be sensitive to asymmetric or over-extended mask regions -- the centerline y-coordinate at each column x is computed as the confidence-weighted mean:

$$y_{\text{hat}}(x) = \frac{\sum_y [y * p(x,y) * \text{mask}(x,y)]}{\sum_y [p(x,y) * \text{mask}(x,y)]}$$

restricted to the column's binary mask extent, with a fallback to the unweighted midpoint where the denominator is degenerate. The resulting curve is passed through an outlier-rejection filter (iteratively reweighted local linear regression, rejecting points deviating from a locally-fit trend beyond a fixed threshold) to remove residual local artifacts.

Columns for which the network produces no confident foreground prediction -- an observed residual failure mode, particularly on images with debris tightly conforming to the shell surface -- are not silently omitted. Instead, the plausible horizontal extent of the shell is separately estimated using a classical Otsu-threshold-based segmentation (used only to bound the column range, not to inform vertical position), and any resulting gap in the network's prediction is bridged: linearly interpolated where bounded on both sides by confident predictions, or extrapolated as a constant value where the gap extends to the boundary of the plausible extent. Each output point retains a boolean flag indicating whether it derives from direct network confidence or from this bridging procedure.

H. Evaluation Protocol

Quantitative evaluation used a symmetric nearest-point (Chamfer) distance between each predicted centerline and its corresponding human-annotated reference curve (the latter resampled via the same spline procedure as Section C), computed independently in pixel units for each of the 107 annotated images using the model checkpoint retained from the final training run (train7.log).

Metric	Value
Images evaluated	107 (0 empty-mask failures)
Mean directed error (pred -> ground truth)	39.74 px
Median directed error	31.12 px
Mean directed error, as % of image diagonal	1.07%
Mean symmetric Chamfer distance	30.56 px
Median symmetric Chamfer distance	23.60 px
Mean non-placeholder (real) coverage	87.4%
Worst-case directed error	223.1 px (debris-fused specimen)
Held-out validation loss (final run)	0.360

The single worst-performing image in the evaluation set exhibited a directed error of 223 pixels, corresponding to a case of debris material fused directly onto the shell surface -- the same failure category motivating the annotation-driven approach -- for which the network produced confident output over only a minority of the image.

Limitation of this protocol, stated explicitly: the reported evaluation set is the full annotated set used across training rounds and is not disjoint from the images on which the retained model was trained (a random 15% subset was held out during each individual training run, but the cumulative evaluation above spans both the final run's training and validation partitions). These figures should therefore be interpreted as a measure of fit to the accumulated human-verified reference set rather than as an estimate of generalization performance to entirely unannotated imagery. The held-out validation loss from the final training run (0.360, reported above) is provided as an auxiliary -- though not directly comparable in scale to the pixel-distance metric -- indicator of generalization. No evaluation against an independent, never-annotated test partition has been performed at the time of writing.

Reference

[1] O. Ronneberger, P. Fischer, and T. Brox, "U-Net: Convolutional Networks for Biomedical Image Segmentation," in Proc. MICCAI, 2015.